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Recovery of rubidium from brine sources utilizing diverse separation technologies

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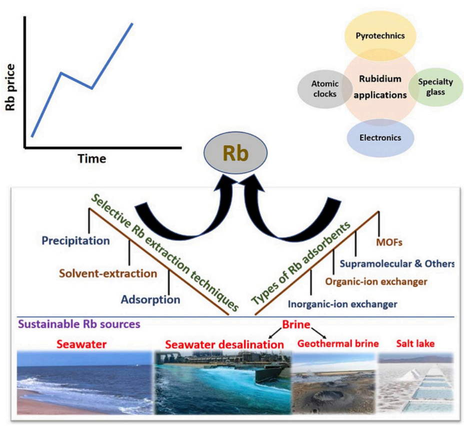
HIGHLIGHTS

- Brine is a sustainable alternative source for Rb recovery.
- Rb application, land reserves, market value and recovery technologies are reviewed.
- Adsorption mechanism for selective Rb recovery from brine is critically examined.
- Economic evaluation of Rb recovery from brine is highlighted.

GRAPHICAL ABSTRACT



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ABSTRACT

A rare alkali metal, rubidium (Rb) has significant economic value and emerging industrial applications including biomedical research, solar cells, atomic clocks, and electronics. Primarily Rb is recovered as an intermediate product during cesium or lithium extraction from pollucite or lepidolite, respectively. The rarity of Rb and its specific industrial usage have necessitated the development of new processes and the identification of alternative sources of Rb. As a result, alternative sources of Rb are becoming more appealing, primarily in the form of seawater brine and salt lakes. Researchers have utilized solvent extraction, precipitation, adsorption, and hybrid membrane-sorption technologies to recover Rb. A more in-depth evaluation of different separation technologies is imperative for achieving selective Rb recovery from complex brines. Hence, this assiduous review focuses on various Rb recovery technologies from brine. A specific emphasis is placed on Rb recovery by ion exchange-adsorption process in view of its efficiency, selectivity, and cost-effectiveness. Efforts to enhance adsorption are also discussed, including metal-organic framework grafting and encapsulation. This review will provide in-

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depth strategies for developing efficient and sustainable pure Rb recovery technologies having maximum adsorption capacity with improved kinetics, re-usability, and easier re-generation of the spent adsorbent from alternative brine sources.

1. Introduction

1.1. Rubidium

Rare alkali metal rubidium (Rb) finds use in biomedical science, specialty glass, atomic clocks, thin-film solar cells, pyrotechnics, and electronics [1–5]. Rb began to attain industrial prominence due to its versatility after 1920. As per reports of the United States Geological Survey, 2020 (USGS 2020), optics, telecommunication systems, and specialty glasses cover the primary market for Rb. A new possible use for Rb is in quantum computing [6].

Rb has a silvery-white and soft physical appearance. Rb's physical and chemical properties are closely akin to those of radio cesium (Cs), which marks it as slightly radioactive. Though, there are no environmental circumstances where Rb is supposed to pose a risk to human life. The electronic configuration of Rb allows it to give away its outer electron easily, which marks it as one of the reactive species in the periodic table [7]. Rb is perceived as the 25th most abundant element on earth's crust and is among the 56 elements that account for 0.05 % of crustal element's weight [8]. Relatively plants respond more rapidly towards the uptake of Rb in the absence of potassium (K^+) ions. In this manner, Rb enters the food chain and therefore contributes to the consumption of 1–5 mg of Rb per day [9].

1.2. Sources of rubidium and extraction processes

i) Land-based mineral ore sources and related extraction processes

The key source for Rb is the land-based mining of ores. Land mining does not yield Rb as an element in its pure metal form. Instead, it is found as a fraction of various ores. It is more abundant in nature than some common elements including lead (Pb), copper (Cu), zinc (Zn), lithium (Li), and Cs [10]. Some independent minerals related to Rb which usually exist in conjunction with rare metals such as beryllium (Be), tantalum (Ta), Cs, niobium (Nb), and Li are zinnwaldite, lepidolite, pollucite, feldspar, and biotite. The relative percentage of Rb occurring in these minerals is listed in Fig. 1 [10–14]. As per USGS 2022, Rb is primarily recovered during the extraction of Cs/Li from pollucite/lepidolite minerals. Lepidolite and pollucite, the prime Rb-containing mineral sources, comprise Rb in the form of oxide. Lepidolite, a lithium-rich mica mineral, contains approximately 3.5 % Rb_2O and pollucite $(Cs, Na)_2Al_2Si_4O_{12}.2H_2O$ containing iron, potassium, and calcium with about 1.4 % Rb_2O [15,16].

Rb is extracted from mineral ores by thermal hydrometallurgy ore roasting and acid leaching methods [17]. Thereafter different post-treatments such as precipitation, crystallization, and solvent extraction methods, are employed to recuperate these ions, as presented in Fig. S1 [18]. The initial phase of the suggested process in Fig. S1 is mechanical activation. The nature of mineral structure changes during the activation process, resulting in amorphization and increased chemical reactivity. Rb and other elements in the mineral are then digested to create soluble sulfates, and Rb can be recovered through crystallization following leaching and roasting. Additionally, repeated crystallization helps to achieve a higher recovery of Rb. To improve Rb yield, precipitation and solvent extraction techniques can be utilized. The roasting process involves adding alkaline-based leaching chemicals to the ores at high temperatures. Recent studies illustrate that alkaline leaching based on autoclaving can be effectively applied to decompose Rb-containing minerals [19].

High grade mineral ores are declining rapidly and compounding it with the fact that the techniques involved in extracting Rb are water as well as energy-intensive creates a grim scenario [20]. Hence, the supply of Rb should not be contingent only on land-based mining but other sustainable sources of Rb like seawater brine, salt lake brine should also be explored.

ii) Other sustainable sources and related extraction processes

Sea water, which covers almost 70 % of the earth's crust, encompasses a considerable quantity of valuable elements including Rb and can serve as an alternative source for mineral recovery. Almost every element enumerated in the periodic table can be found in seawater, either in ample quantities or in a lower concentration. Depending on the concentrations, these elements can be categorized into "major elements" having a concentration of (10^4–10^6 mg/L) and "minor elements" having a concentration of (10^{-4}-10^0 mg/L).

As per the global data, around 84 % of total desalination plants operating worldwide are based on reverse osmosis which marks it certainly the most dominant desalination process [21]. Primarily, all the desalination processes are operated at a recovery ratio of about 45–50 %, which sources the massive generation (124–130 million cubic meters

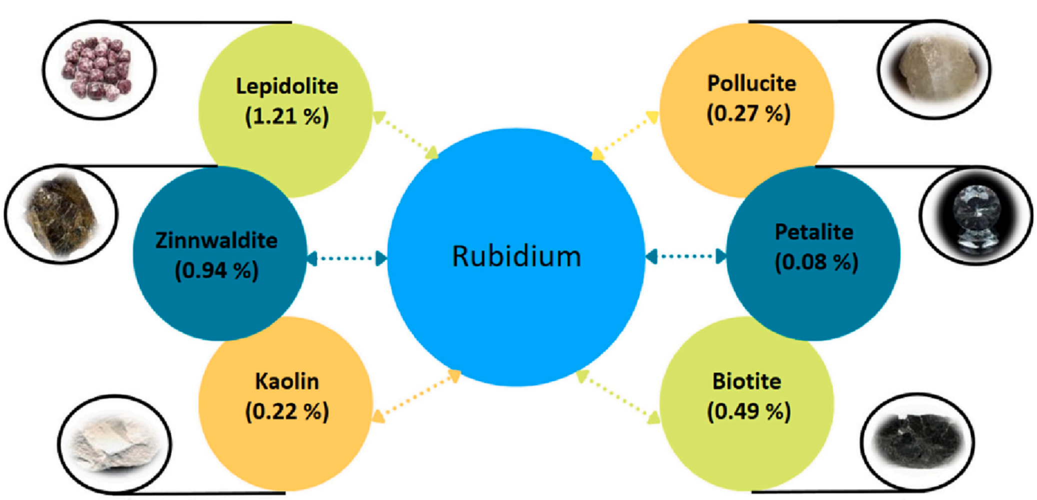


Fig. 1. The figure depicts the relative percentage of rubidium occurring in independent minerals such as Zinnwaldite, Lepidolite, Pollucite, Petalite, Biotite, and Kaolin.

per day) of concentrated brine globally [14,17,18]. Brine from seawater contains high concentrations of total dissolved solids (TDS) and has numerous valuable elements, salts, and ions that can be recovered and ultimately cut down on waste disposal costs [22]. Rb is one of the valuable elements present in brine with an average concentration lying in the range of 2–10 mg/L [22,23]. However, the relative abundance of Rb also relies on the type of brine. More discussion of brine as an alternative source for Rb recovery is discussed in chapter 3.

A variety of Rb removal methods from brine have been employed and applied for practical purposes including precipitation, solvent extraction, adsorption, membrane separation, and hybrid methods [21,24,25]. Recent scientific articles on Rb recovery have emphasized adsorption more. Fig. 2 represents the proportion of scientific papers discussing different Rb recovery methods. Multiple types of adsorbents have been utilized for Rb including inorganic ion-exchange adsorbents, organic ion-exchange adsorbents, graphene-based adsorbents, polymer composite adsorbents, and metal-organic-frameworks (MOFs) [26–29]. A detailed analysis regarding the adsorption of Rb utilizing the above listed adsorbent is done in the subsequent chapters of this review.

Previous reviews have explored Rb recovery, limited to land-based mining only. Xing et al., in their review, primarily addressed the occurrence of Rb in mineral sources, progress in extraction processes, and mechanism involved in the Rb recovery from minerals. Also, the review holds a brief discussion on extracting Rb from brine sources [30]. Zhang et al. and Gao et al. discussed the conventional methods (precipitation and solvent extraction) for recovering Rb and Cs from brine [31,32]. It has been discovered that there is a lack of thorough analysis for Rb recovery from natural brine in past review attempts. Before going into the recovery processes of Rb, it is also essential to understand the importance of Rb as an element. We believe other aspects of Rb (properties, application areas, sources available, and the global presence of Rb) provide holistic knowledge and will help in understanding the Rb market value, industrial implications, supply, sources, and demand. Also, there is still a gap in the specific role of adsorption processes for selective recovery of Rb from complex brine sources.

This review explores the general context of the industrial application of Rb and its expanding market, making it crucial to maintain the balance between supply and demand. Also, the challenges associated with land-based mining of Rb as a complex and chemically intensive process to achieve pure Rb salt are discussed, followed by a detailed compilation of the characteristics of various alternative natural brine sources for Rb recovery across the globe. Thereafter, this review provides a technical discussion on the various methods available for Rb recovery from brine. A more focused technical emphasis of this review is on adsorption methods utilized for Rb recovery from various brine sources. The mechanism involved in the recovery process, the effect of co-existing ions, and the effect of different operating parameters on the Rb recovery efficiency are conferred. New developments in adsorbents, including the application of metal-organic framework is discussed. Additionally, an economic outlook along with the future strategy for Rb recovery is highlighted.

2. Importance of rubidium

2.1. Application status of rubidium

There has been a significant advancement in the use of Rb since its discovery [33]. Formerly this metal was solely momentous for research and development purposes, particularly in the domains of electronics and chemistry [34]. Due to technological improvements, it is currently utilized in several diverse industrial sectors. The mounting interest in biomedical research may be one of the principal factors driving global demand for Rb [35]. The development of a digital X-ray intensifier, single-crystal detection, and several new high-tech devices has elevated the use of this element and its related compounds in a highly technological field [5]. Special glasses with rubidium carbonate (Rb_2CO_3) are used in telecommunication systems and night-vision devices [36]. Numerous detection methods utilize Rb in combination with a semi-metal element as it possesses light-emitting capabilities [37]. Rubidium tellurium (Rb_2Te) is applied in photoelectric cells to offer a photoelectron emitting surface. In addition to usage in medical imaging equipment, a rubidium antimony cesium coating is frequently applied over a photo cathode [34]. Rb is also employed as sedatives, soporifics, and anti-shocking agents in a number of medico scientific applications. Salts such as chlorides of Rb are utilized as density gradient medium for centrifugal separation of viruses and sometimes for DNA and RNA [38]. Numerous atomic studies have preferred Rb atoms because of their laser cooling effect, and there has been a recent increase in demand for Rb in this field [39]. Atomic clocks are requisite in satellite-based navigation systems, and Rb atomic frequency clocks have proven helpful [40]. Fig. 3 highlights significant percentage values for each Rb usage. Uses and application potential for Rb are significantly expanding as research advances over the years. Numerous usages of Rb fall within the minor application category illustrated in supplementary information [34].

2.2. Worldwide rubidium minerals and market value

Substantial Rb-bearing pegmatite has been recognized globally in Australia, Canada, Denmark, Germany, Peru, and Russia [36,37]. Although, minor amounts have also been testified in brines of China, Chile, and some parts of South Asia [42]. Despite the fact that there are no official records concerning Rb production, it is known to occur in Canada, Zimbabwe, and Namibia. North America is considered to be the largest market for Rb, and this growth is likely to be augmented because of elevating Rb industrial applications. Some occurrence of Rb demand is known in Arizona, Alaska, Idaho, Utah, and South Dakota of the United States (US), and the dominant supply is from Canada [43]. Table 1 entails the presence of Rb across the globe. According to reports from US industry data, the domestic consumption of Rb has been to the tune of 2000 kg per year in the last decade, about 80 % of which is utilized in the high-tech industry. The remaining 20 % is employed for

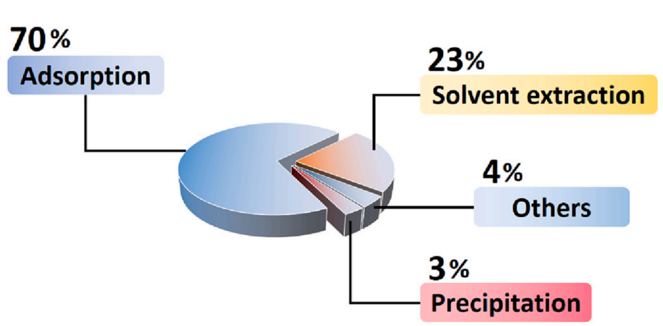


Fig. 2. The figure shows the number of published research articles based on different Rb recovery processes such as adsorption, precipitation, solvent extraction, and other methods.

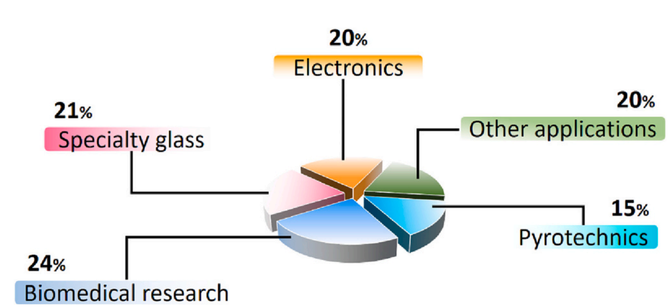


Fig. 3. The figure shows the relative percentage share of various industrial use of Rb. Biomedical research-24 %, Electronics-20 %, Specialty glass-21 %, Pyrotechnics-15 %, Other applications-20 % [41].

Table 1
Rubidium presence globally categorized as major and minor reserves of Rb [45].

Countries with major Rb reserves	Countries with minor Rb reserves
United States, Peru, Canada, United Kingdom, Germany, Denmark, Afghanistan, Kazakhstan, Russia, Japan, Australia, Zimbabwe, Namibia, Zambia	China, France, Chile, evaporites of United States (New Mexico and Utah)

existing use [44]. In addition to the US, countries such as Japan, Europe, Russia, and other Organisation for Economic Co-operation and Development (OECD) countries are involved in the application of this element [45]. Table S1 shows the estimates of the Rb reserves globally.

Currently, the global Rb market is estimated at 2.57 kt and is expected to grow by 4.01 % from 2022 to 2027 [44]. The average cost of Rb is currently USD 14720 per kg (1.1 million INR). The price is expected to increase, predominantly due to the influence of cost and the complexity of recovering Rb from rock ores. Fig. 4 reflects the increment in the price of Rb per gram for the US (from 1990 to 2020). This graphic makes it evident that the trend is upward and accelerating.

3. Brine as an alternative source for rubidium recovery

Rb recovery requires several intricate steps, making the recovery of pure Rb from land mining challenging and unsustainable in terms of chemical requirement, process intensity, and economic cost. Thus, the difficulty of Rb land mining has motivated Rb recovery from alternative water sources, such as salt lakes, seawater, and its related brine. The development of technologies to recover minerals from brines, including Rb, not only contributes to reducing emissions to the environment but also takes advantage of these abundant mineral resources, reducing dependence on land mining and ores [23,46–48].

3.1. Seawater desalination brine

The presence of Rb in natural seawater is well established by previous sample analysis. The concentration of Rb in seawater and its related seawater brine (0.18–0.20 mg/L), alike Li, is significantly lower

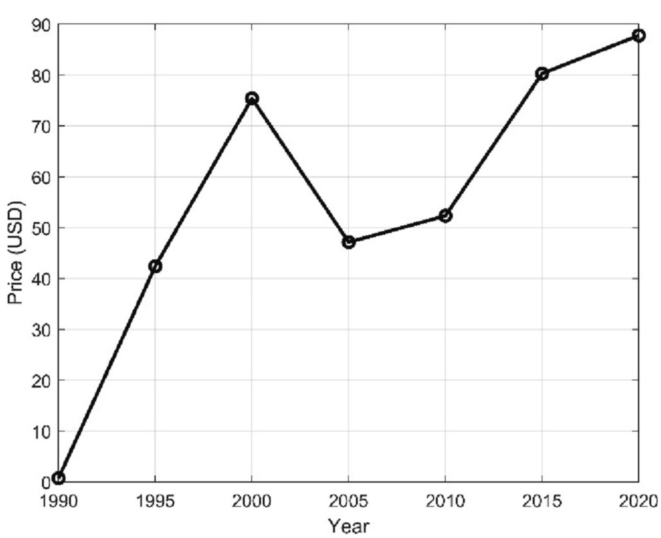


Fig. 4. Price of Rb/g (99.75 %–99.8 % purity) in the United States (for the period of 1990–2020) is reflected. There is a rising trend in the price of Rb from 1990 to 2000. However, a dip can be observed during 2000–2010. This decline in value can be explained by the fact that the decade was known as the recessionary decade. Although with increasing application potential in biomedical research, atomic clocks, solar cells, etc., the market price of Rb elevated from 2010 to 2020. The market is expected to grow during 2022–2027 [45].

than predominant ions, specifically sodium (Na), potassium (K), magnesium (Mg), and calcium (Ca) as illustrated in Table 2 [49].

Although Rb concentration is fairly low in seawater, due to the extensive and renewable amount of seawater present globally, the total amount of Rb in seawater ($\sim 2.5 \times 10^{11}$ t) exceeds that of land ores [50]. Moreover, in a desalination plant, recovering economically valuable Rb from seawater brine before being directed back to the ocean may provide additional revenue to offset brine treatment costs [51]. It must be acknowledged that the revenue offset prospects may only be possible if the proposed concept is cost effectively profitable based on future projected demand for Rb, given that the current Rb demand is still relatively low. In essence, the alternative solution of using the concentrated brine resourcefully to recover Rb, rather than simply disposing of it, is anticipated to achieve both environmental and economic benefits.

Desalination plants operate in over 120 countries, primarily in the US, Gulf region, regions of Europe, Australia, and Asia. Globally, it is appraised that over 100 million m^3/day of fresh water is produced from seawater in reverse osmosis desalination plants [52]. On an average, if it assumed that these desalination plants operate at a capacity of 50 % reverse osmosis rate, these plants would produce around 100 million m^3/day of brine. In essence, if we assume that the recovery efficiency is 98–100 %, a mass of 15–20,000 kg Rb per day (at an average concentration of 0.15–0.20 mg Rb/L in brine) can be extracted from these brines directly through the outlet discharge pipeline. The main factor determining the economic viability of Rb recovery from a desalination plant is the cost-effectiveness of the method employed to achieve selective Rb recovery. In the following chapters of this study, several selective recovery strategies are covered in further detail.

3.2. Geothermal brine

Geothermal brine has been well acknowledged as a highly potential alternative source for resource recovery, given the natural presence of valuable metals in geothermal brine. Unlike the vast and uniform distribution of seawater and its related desalination brine as detailed in Table 2, geothermal related brine is only relevant to countries that utilize geothermal power/energy and hence produce the brine, predominantly North America, Asia (Philippines, Indonesia, Japan, China), New Zealand, Mexico, Russia, and Iceland [56]. Further, the distribution and composition of valuable metals in geothermal brine can vary significantly within a geographical location, as illustrated in Table 3. Comparatively, the concentration of Rb in geothermal brine differs considerably in different wells. In many cases, Cs is also present alongside Rb in geothermal brine, unlike seawater brine. Effectively recovering Rb over Cs from geothermal brine may be difficult because of the close similarities in their properties. This apart, the naturally heated condition and acidic pH of the brine may further hinder Rb recovery.

3.3. Salt lake

Rb concentrations in salt lake brine range from 169 mg/L to 290 mg/L as mentioned in Table 4, prompting multiple studies to investigate the possibility of recovering Rb from salt lake brine. In most instances, the presence of competitive Cs is minimal in salt lake brine. Nevertheless, natural salt lake water with substantial Rb concentration is primarily located around China, Israel, and the US [32,62,63].

4. Selective extraction techniques

4.1. Precipitation

Chemical precipitation, a well-entrenched conventional method, has been applied regularly to exclude radionuclides. This technique is worthwhile due to its simplicity, ease of use, and suitability for large-scale applications. Additionally, this method can be utilized in conjunction with other techniques including membrane separation and

Table 2
Concentration of Rb and major ions present in natural seawater and seawater brine from desalination plants.

Brine source	Ion/metal concentration (*mg/L*)								Ref.
	Na	Mg	K	Ca	Sr	Rb	Li	Cs	
Seawater	10,764	1277	398	412	7.88	0.12	0.173	<0.001	[49]
	11,393	1414	20.31	416.81	7.85	0.13	0.18	–	[53]
Seawater brine									
El Prat de Llobregat (Spain)	27,521	2450	554	–	–	0.19	0.27	<0.001	[54]
La Skhira (Tunisia)	10,500	1350	380	–	–	0.12	–	<0.001	[55]
Perth (Australia)	22,100	2570	783	894	15.40	0.20	0.40	–	[51]

Table 3
Concentration of Rb and major ions present in geothermal brine.

Geothermal brine	Ion/metal concentration (*mg/L*)								Ref.
	Na	Mg	K	Ca	Sr	Rb	Li	Cs	
Salton Sea (USA)	53,000	33	16,700	27,400	411	170	194	20	[57]
Brawley (USA)	47,600	114	12,600	21,500	1043	67	219	19	[58]
Wairakei(New Zealand)	1250	0.04	210	12	–	2.90	13.2	2.50	[59]
Salak (Indonesia)	5000	0.10	990	320	4.50	5	17	4.50	[58]
Cerro Prieto (Mexico)	8300	0.50	2210	521	16	11	27	39	[60]
Russia									
Hvergerdi (Iceland)	212	–	27	1.50	–	0.04	0.30	<0.02	[59]
Mote Amiata (Italy)	1977	<0.50	558	128	2.40	2.10	21.90	0.70	[61]

Table 4
Concentration of Rb and major ions present in salt lake brine.

Salt lake brine	Ion/metal concentration (*mg/L*)								Ref.
	Na	Mg	K	Ca	Sr	Rb	Li	Cs	
East Taijinar (China)	24,850	40,640	15,630	–	–	18.50	1020	1.30	[62]
Zabuye (China)	129,600	–	40,900	–	–	50–60	1050	12–21	[32,63]
Khag Nuur (Mongolia)	391	145	50	50	3.50	1.35	0.03	–	[64]
Dead Sea (Israel)	23,968	39,600	6670	15,720	297.90	60	–	–	[63,65]
Salton lake (USA)	12,370	1400	258	944	22	137–169	–	16–20	[32,66]
East Transbaikalia (Russia)	125,000	400	440	80	9.20	0.05	0.64	–	[67]
Vestfold Hills (Antarctica)	38,800	3990	1230	810	10.90	0.34	–	<0.01	[68]

evaporation for enhanced recovery and purification. Nevertheless, the process is limited due to the generation of secondary products and the difficulty in selectively separating ions [32]. During precipitation, the conflicting ions in brine sources evoke problems. These ions should be eliminated first using precipitation and centrifugation [31]. Furthermore, the filtration process and decantation are necessary for selective Rb separation. Membrane technology can provide adequate filtration to concentrate the brine, thereby improving recovery efficiency while producing fresh water [69]. Further, filtration through a membrane can be accomplished with evaporation for greater recovery and purification. Finally, the flotation method can concentrate the Rb^+ [70]. Choosing a suitable precipitator for Rb recovery depends upon the solubility constant (Ksp) values. Smaller values of Ksp are favored for optimal precipitation [71]. The values of solubility constant for various Rb compounds are listed in Table 5. Practically, Ksp values lower than 10^{-10} are more benign in chemical precipitation. As can be seen from Table 5, the Ksp values of various Rb compounds are in this range.

Commonly utilized precipitants for Rb are tin tetrachloride ($SnCl_4$), aluminium sulphate ($Al_2[SO_4]_3$), silicon-tungsten acid ($H_4[SiW_{12}O_{40}]$), oxalic acid ($C_2H_2O_4$), sodium iodide, and potassium iodide (NaI, KI) [42,67,72]. Additionally, Rb has been eliminated by precipitation using the unique salt class known as heteropolyrates. In this reference, ammonium phosphowolframate shed promising exchange capacity for the precipitation of Rb^+ [70]. Its regular crystal appearance, symmetric structure, smooth surface crystal along with high selectivity makes it suitable for precipitation. However, these usually arise in fine powder form which confines its usage as it becomes challenging to separate the precipitates. In order to resolve this issue, efficient and fast approaches for solid isolation can be employed. Some of the traditional separation techniques utilized for this purpose are evaporation, decantation, membrane filtration, and centrifugation which are shown in Fig. 5 [73]. However, the Rb precipitation process is affected by the presence of K^+ and decreases with higher K^+ concentrations [74].

Table 5
Values of the solubility product constant for rubidium ion compounds [75].

Compound	Chemical formulae	K_{sp}	pK_{sp}
Rubidium cobaltinitrite	$Rb_2[Co(N0_2)_6]$	1.5×10^{-15}	14.83
Rubidium hexachloroplatinate	$Rb_2[PtCl_6]$	6.3×10^{-8}	7.20
Rubidium fluoroplatinate	$Rb_2[PtF_6]$	7.7×10^{-7}	6.12
Rubidium hexafluorosilicate	$Rb_2[SiF_6]$	5.0×10^{-7}	6.30
Rubidium perchlorate	$RbClO_4$	3.0×10^{-3}	2.52
Rubidium peridodate	$RbIO_4$	5.5×10^{-4}	3.26

4.2. Solvent extraction

Solvent extraction is an extraction process based primarily on the relative solubility of the target material under two different immiscible liquids. It is accredited for its high selectivity and operation convenience however it involves the generation of secondary organic pollutants that are highly toxic [76]. The majority of the Rb^+ is extracted into the organic phase through solvent extraction, then back extracted. The two most often used extractants for solvent extraction are phenolic extractants and crown ethers [33]. They have a superior recovery efficiency for Rb. Detailed discussion on extractants and solvent systems used for Rb recovery is given in subsequent sections of this chapter.

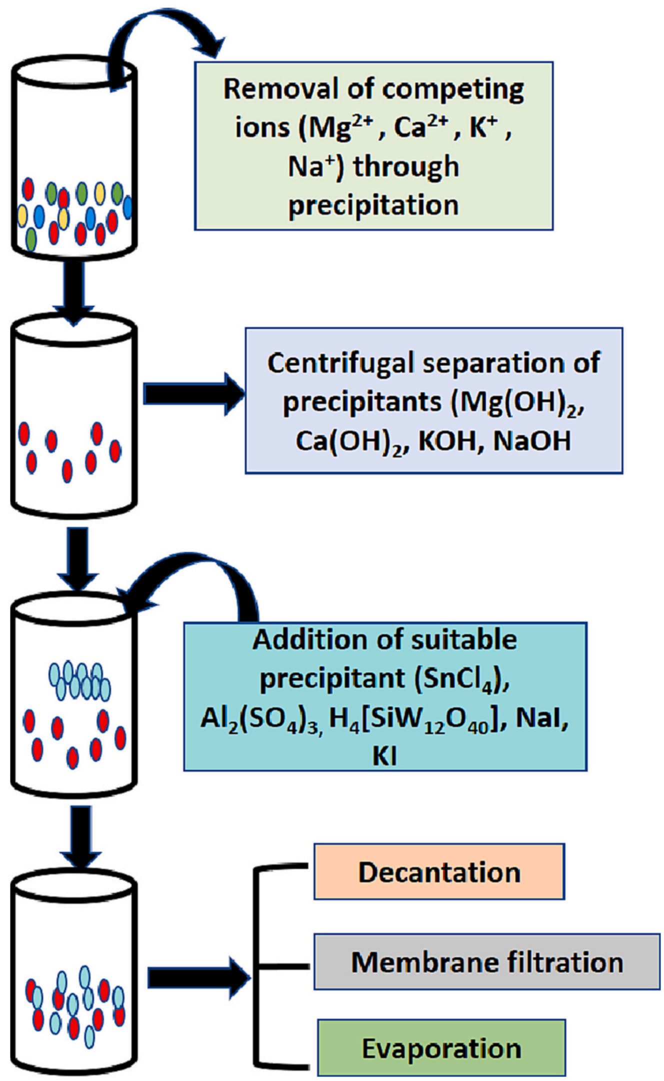


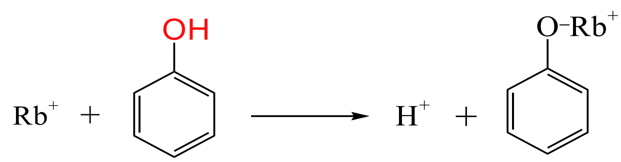
Fig. 5. Sequential flow chart for various processes (The figure shows the different processes employed with the precipitation method for enhancing the recovery and purification of Rb ions).

4.2.1. Extractants

Various sorts of extractants have been extensively reported for the removal of Rb^+ recently, including sodium triphenylcyanoboron (STPB), phenols, substituted phenols, and crown ethers [16,45]. Overall, the coordination of the extraction process depends in large part on structural alterations brought by various solvent systems [77]. Of all these listed extracts, substituted phenols, primarily 4-tert-butyl-2-(@-methylbenzyl) phenol (t-BAMBP), is efficient and has been broadly studied [78,79]. On the other hand, STPB has also shown satisfactory results. STPB is composed of three phenolic rings bonded to boron and is a white crystalline solid, as shown in Fig. 6a. This compound is utilized as a precipitating agent in inorganic and organometallic chemistry for NH_4, K, Rb, and Cs [76]. Rb is extracted as ion or ion-dissociate using STPB. This is because STPB creates an anion metal complex coupled with cations and has shown selectivity for Rb among different salts of its category. Alkali metals, particularly large alkali metals, have a chemistry that inclines to prevent the formation of the extractable molecule. Since large alkali metals are prone to form cations, their cations are unstable due to lower solvation enthalpies/energy. As a consequence, it is difficult to establish a completely stable extractable molecule. Due to this, alkali metals having large ionic radius can only be extracted as ionic pairs with very large ions [80]. The introduction of organic solvents having higher dielectric constant and water-immiscibility allows the best possible extraction. O-nitrotoulene as the solvent for STPB has demonstrated greater stability and phase separation.

Another class of Rb extractants comprises crown ethers. These are coronal compounds having holes in their structure [81]. Mostly these are apt under highly acidic conditions [82]. They undergo complexation reactions when reacted with cation or alkali metal ions. The formation of the most stable complex is favored when the size of the cavity is similar to the radius of the alkali metal cation. 18-crown-6, benzo crown ether, and dicyclohexane-18-crown-6, as shown in Fig. 6b, are the most extensively applied and studied crown ethers for the recovery of Rb. The efficacy of crown ether to remove Rb from brine is also evaluated by examining the metal ion distribution ratios [83,84]. It is observed that K^+ has a serious effect on the recovery of Rb^+ [83,85]. Hence it becomes crucial to inspect the selectivity of Rb by crown ethers in the presence of these ions. Calixarene, a versatile macrocycle, has also been considered an extractant [86]. Ionizable calixarene is more effective in cationic degradation than non-ionizable calixarene [87].

BAMBP and its acidic substituted version t-BAMBP are Rb's most extensively used extractants. This is due to their remarkable selectivity, low water solubility, high stability, non-volatility, ease of stripping, low cost, and other advantages [80]. The improved separation coefficient values exhibited by t-BAMBP make it a better extractant for Rb [88]. The underlying mechanism for the recovery of Rb^+ using phenol extractants is the exchange of H^+ with Rb^+ under alkaline conditions, followed by the formation of hydrophobic salts of phenol in the organic phase. The following equation can be considered to understand the mechanism of this process.

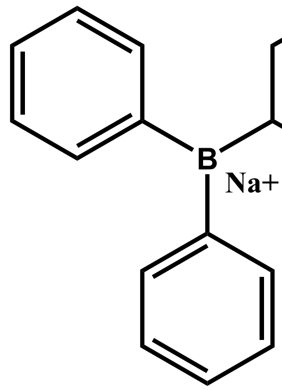


(1)

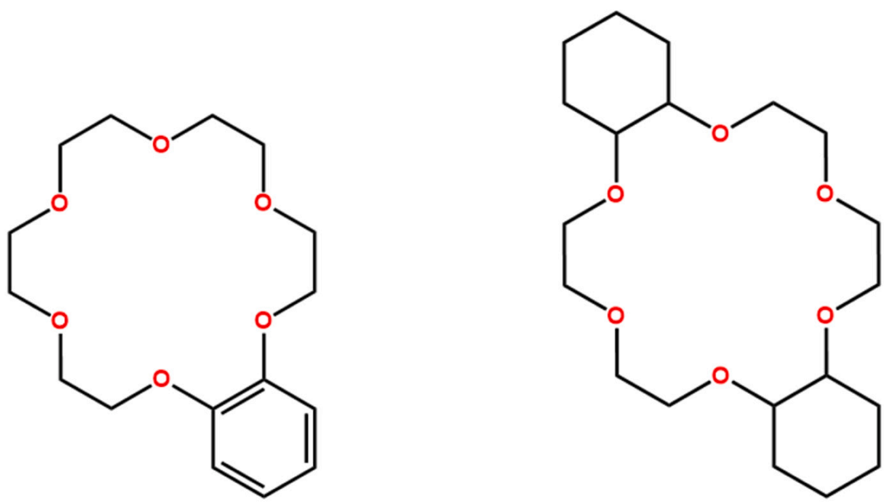
t-BAMBP has similar extraction behavior as compared to other forms (s-BAMBP), but they are less challenging to synthesize [89]. The affinity for extracting different alkali metals and utilizing them is such that it is maximum for Cs and minimum for Li following the order as Cs > Rb > K > Na > Li. t-BAMBP, when utilized as a single entity, causes viscosity and hence makes phase splitting and emulsification challenging. This issue can be resolved by the application of a diluent that serves as the carrier solvent for extracted complex [79]. In this reference, Liu et al. studied the Rb recovery using t-BAMBP along with sulfonated kerosene as a solvent [90]. The procedure resulted in more efficacy and can be used to extract Rb and Cs simultaneously. Table 6 represents the efficiency of Rb recovery utilizing various extractants and diluents.

4.2.2. Solvent systems

Alkali metal extraction depends on solvation, polarity, and solubility. Various solvent systems have been utilized, including o-nitrotoulene, ketone, chloroform, kerosene, cyclohexane, dodecane, and 2,6 dimethyl-4-heptanone [78,91]. Out of them, o-nitrotoulene has revealed satisfactory results because of its high dielectric constant value that offers the dissociation of ion pairs in organic liquids. For practical utilization of crown ether, it is significant for a diluent to dissolve the precipitate containing Rb. As a diluent, nitrobenzene has been effective, but it is usually unacceptable in many countries because it has dangerous characteristics. Non-toxic aromatic hydrocarbons can replace nitrobenzene, but they work well in an alkaline medium [92,93]. Regarding the environmental degradation and toxicity values imposed by the volatile organic solvents, room temperature ionic liquids (RTILs) have been utilized and are considered more environmentally benign [94]. They are promising solvents because of their low melting temperatures, low volatility, and short phase separation times.



a. Sodium tri phenyl boron (STPB)



b. (a) Benzo 18-crown-6   (b) Dicyclohexane-18-crown-6

Fig. 6. a. Sodium tri phenyl boron (STPB).
b. (a) Benzo 18-crown-6 (b) Dicyclohexane-18-crown-6.

Table 6
Rubidium recovery results utilizing numerous extraction systems as per recent studies.

Rubidium source	Extractant	Diluent	Extraction efficiency (%)	Ref.
Simulated brine	18C6	[C4mim][PF_6]	84.11	[95]
Simulated salt lake brine	t-BAMBP	Cyclohexane		[79]
Salt lake brine	t-BAMBP	dodecane	86.50	[96]
Seawater	t-BAMBP	Kerosene	98	[80]
Potassium-rubidium old brine	t-BAMBP	Sulphonated kerosene	91 %	[88]
Aqueous solution	BPC6	C_2mimNTf$_2$	~100	[94]
Simulated brine	D24C8	Nitrobenzene	99	[91]
Synthetic brine solution	t-BAMBP		99	[89]
Salt lake brine	t-BAMBP	Kerosene	95.04	[90]
Salt lake brine	t-BAMBP	Sulphonated kerosene	96	[97]
Potassium-rubidium old brine	Phenol		73	[98]

4.3. Adsorption

The approaches defined above for Rb recovery are constrained in several ways. The process of precipitation at times creates secondary pollution and their removal leads to higher energy consumption. This process also entails added flocculation systems to perform auxiliary precipitations. Solvent extractors become less efficient when operated with a wide pH range. Their effectiveness is likewise impacted by the influence of competing ions. As brine contains a variety of different elements that interfere during the recovery of Rb^+, it becomes grim to maintain the selectivity and efficiency for recovery using these processes [54]. Compared with other methods, the adsorption process is highly flexible to attain selective recovery of Rb^+ primarily through physical and chemical manipulation of the ionic size. This process performs superior when exposed to diverse salt levels and is suitable to operate in a variety of pH situations [99]. Fig. 7 defines the percentage of work done utilizing different adsorbents. The adsorption process accounts for a significant proportion of all research done regard to selective Rb recovery.

Adsorption takes into account the characteristics of the target ions and solutions. Several aspects should be taken into account when

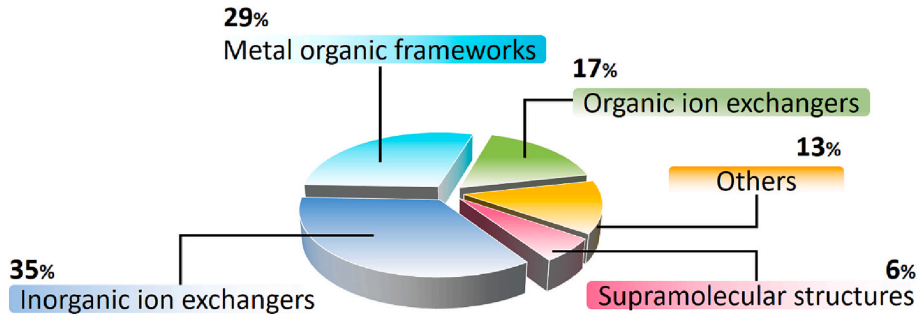


Fig. 7. Percentage of work done utilizing different adsorbents. This categorization is as per literature available on different kinds of adsorbents used for Rb adsorption from brine and aqueous water solution available at google scholar, science direct, and web of science from 1970 to 2022. A list of the categorized publications for each category is provided in the supplementary information.

choosing the adsorbents. These comprise physical parameters such as specific surface area, pore volume, porosity, and chemical parameters, including surface charge, polarity, and the presence of functional groups. An ideal adsorbent should have additional features, including ease of use, selectivity, and cost-effectiveness. Apart from this, adsorbent characterization, and adsorption experiments containing isotherms, kinetics, and thermodynamics studies are obligatory to understand the mechanism behind the adsorption process.

4.3.1. Inorganic ion-exchange adsorbents

Inorganic ion exchange adsorbents are cation-based exchanges that work primarily by exchanging positively charged ions from the adsorbent with a specific ion from an aqueous stream. The ion is then recovered from the ion exchange adsorbent by chemical regeneration as shown in Fig. 8a. General features and characteristic of a cation exchange adsorbent that enables the recovery of a specific target cation are (i) its negative surface charge that enables cation surface attraction, physical adsorption and attachment onto the adsorbent surface; (ii) the presence of pore channels on the adsorbent that enables ion diffusion through the pores (bulk, film and intraparticle diffusion); (iii) structural shape for capturing the target ion, and (iv) ion/metal contents and presence of functional groups within the adsorbent that enables inter ion exchange as depicted in Fig. 8b. For the specific case of Rb, the capacity of an ion exchange adsorbent to attain selective Rb recovery in a mixed saline brine is highly dependent on the (i) capture structure of the adsorbent, (ii) the close match of the ionic radius and (iii) matching ionic charge of Rb with the ion/metals being exchanged.

Based on the ionic size of the monovalent alkali metals, it can be evidently noted from Table 7 that exchange between K and Rb can attain a selective capacity of alkali Rb sorption given the closely similar characteristic between them. On the other hand, the primary competitive ion for Rb would be Cs due to the matching hydrated ionic radius of Cs and Rb. Hence, selective recovery of Rb in the presence of Cs would be a challenge, especially for brines containing a high concentration of Cs over Rb. On another note, the adsorbent must also perform well in the presence of high Na concentration, given that most brine solutions are highly saline. This is discussed in detail in the following section.

Table 7
Hydrated and unhydrated ionic radius of monovalent alkali.

Alkali metal	Hydrated ionic radius, Å	Unhydrated ionic radius, Å
Cs	2.26–2.28	1.68
Rb	2.26–2.29	1.48
K	2.32–3.31	1.33
Na	2.76–3.60	0.95
Li	3.40–4.70	0.60

4.3.1.1. Relationship between adsorbent structure and Rb capture capability. i. Zeolite

Natural zeolite (from various origins or different monocationic forms) is a widely used ion exchange adsorbent because of its low material cost, natural presence around the world, commercial availability, diverse structure, and composition. The structure and composition of crystalline microporous aluminosilicate zeolite is promising as an ion exchange adsorbent. This is due to their framework structure being tetrahedral anionic with enclosing cavities occupied by charge-balancing cations and water molecules. Both have enough freedom of

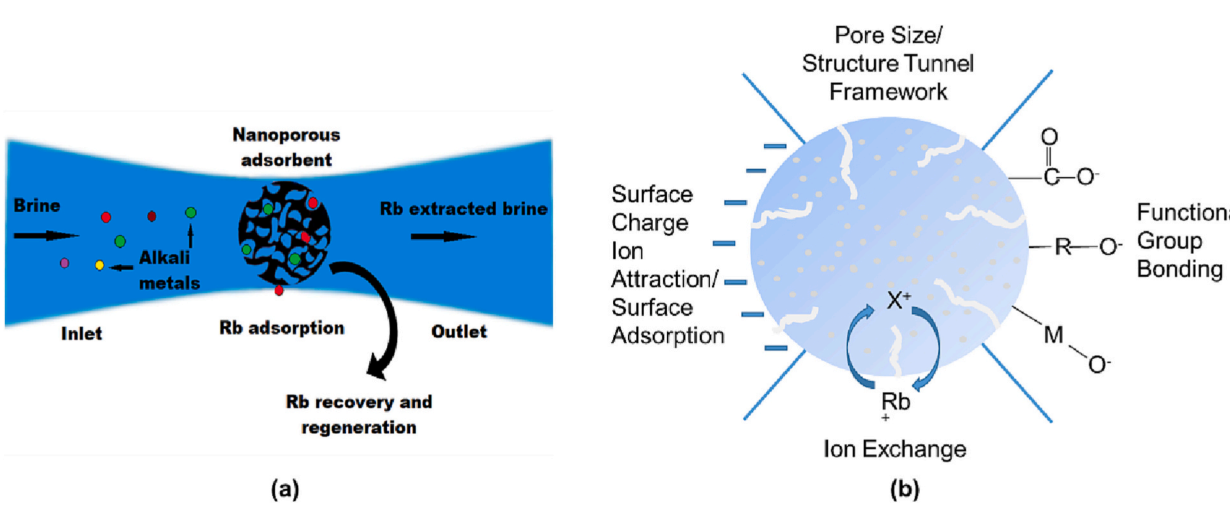


Fig. 8. Representation of the (a) main features and (b) mechanism of an ion exchange adsorbent for selective Rb recovery.

movement to permit cation exchange, as shown in Fig. 9. Specifically, the cavity diameter size of different zeolite types for capturing Cs has been well established. For instance, mordenite zeolite with 12-ring cavities in its structure has a large diameter of approximately 6.50 Å. Likewise, apertures in the chabazite 8-ring cavities have a diameter of 3.80 Å, and as such, Cs (2.80 Å) can easily access and fit into the cavities of these zeolite structures [100]. Although the application of zeolite for Rb capture has not been widely explored. However, given the similar ionic radius of Cs and Rb, zeolite can be expected to efficiently capture Rb as well.

Nevertheless, it must be noted that the sieving and cage capturing mechanism based on the structure of zeolite may not entirely achieve a high selective capacity of a target alkali ion in the presence of other alkali ions. Given that the ionic radius of most of the monovalent alkali ions falls in the range of 2.50–3.50 Å, therefore they can access and enter the inner cavity of zeolites. As such, the selectivity would be highly dependent on the zeolites ion exchange mechanism. This is discussed in Section 4.3.1.2.

ii. Titanium silicate

The crystal structure framework of microporous titanium silicate is rather unique as it has two distinct tunnel sites, with each cell unit consisting of four molecules [101,102]. Another unique feature of the titanium silicate structure is that it captures the unhydrated alkali ions within its structure [103]. Based on the ionic radius of unhydrated alkali ions, Li and Na have the smallest unhydrated ionic radius. Hence, they could occupy the entire framework and the inner/center of the tunnel. Meanwhile, larger alkali ions, such as K, Rb, and Cs, are unable to fit into the inner framework of titanium silicate and therefore, would only be able to occupy available vacant sites in the outer framework. Based on the structural capacity of titanium silicate, in a mixed brine containing all alkali ions, the affinity for ion capture will be in the order of Li > Na > K > Rb > Cs [104]. For instance, Clearfield et al. [101] examined the sorption of titanium silicate towards several alkali ions (Li, K, Na, Rb, Cs). Although titanium silicate material was found to have a high sorption capacity for Rb (5.60 meq/g, equivalent to 478.80 mg/g), the selectivity was relatively low compared to Li, Na, and K. Due to the large hydrated radius, Rb only occupied tunnel sites and was inaccessible to the framework sites of the crystal structure of titanium silicate. On the other hand, based on the smaller size, Li^+ and Na^+ reached these active zones of the sorbent. In practice, it is not feasible to apply this material for selective Rb recovery from brine as the presence of competing cations with concentrations many times higher than Rb is a major obstacle.

iii. Zirconium phosphate

A number of different zirconium phosphate structures, including crystalline and semi-crystalline α type, amorphous as well as dehydrated γ type, have been investigated for alkali metal capture. The general structure of zirconium phosphate is a two-dimensional layered structure, as shown in the Fig. 9c. Within each layer, the zirconium atoms are octahedrally coordinated by four oxygen atoms of phosphate groups located within the layer and two oxygen atoms of the dihydrogen phosphate group located at the edge of the layer. Further, the tunnel bridge between dihydrogen phosphate and zirconium results in cavities/channels. These channels create free diffusion of incoming cations. These materials have a high network window size that allows Rb and other alkali ions in the brine matrix to penetrate the structure of the sorbent [54]. One advantage of zirconium phosphate is that the structure interlayer spacing and cavity size can be flexibly altered based on the target alkali ion which is to be captured. For instance, Cheng et al. [105] used sodium fluoride as a mineralizer that enhanced the crystalline structure and it formed micro-pores of different shapes. This crystallinity condition removed Cs by over 98 % in the presence of high Na. Nevertheless, one of the main challenges of using zirconium phosphate ion exchange adsorbent is that the base materials, zirconium and phosphate are costly as well as environmentally toxic.

iv. Potassium Metal Hexacyanoferrate

The application of metal hexacyanoferrate ion exchange adsorbent for selective Rb and Cs uptake has been well recognized. One of the chief reasons for this is the structure of this ion exchange adsorbent. Specifically, metal hexacyanoferrate is a cubic crystal structure that contains Co and Fe in the structure lattice and is held together through the cyano functional group. The presence of Fe and Co act like tunnel bridges with the cyano functional groups providing the negative charge balance to the ion exchange adsorbent. The inner cavity/body center of the lattice structure contains K, which acts as the ion exchange metal. At the same time, Co and Fe are transitional metal cations that do not participate in the alkali metal exchange. A previous study by Lehto et al. [106] associated the matching size (1.47 Å) of the inner lattice of the cavities of metal hexacyanoferrate ion exchange adsorbent to that of unhydrated Rb radius (1.48 Å) that allowed for high diffusion entry of Rb into the structure lattice compared to other alkali metals.

4.3.1.2. Relationship between selective Rb recovery and ion exchange mechanism. For the specific case of Rb, the capacity of an ion exchange adsorbent to attain selective Rb recovery over other competitive ions in a mixed saline brine is highly dependent on the close match of the ionic radius and matching ionic charge of Rb with the ion/metals exchanged.

i. Sodium- Rb exchange

The sodium-Rb^+ exchange mechanism is especially reflected by sodium based on titanium silicate and sodium based zeolite. Clearfield et al. [104] reported that crystalline sodium titanium silicate material has two exchange sites in the structure, half the amount of Na^+ distributed in the framework and others within the tunnels. As mentioned above, due to the sizeable unhydrated radius, Rb^+ could only replace Na^+ in the centre of the tunnel if it is present without any other

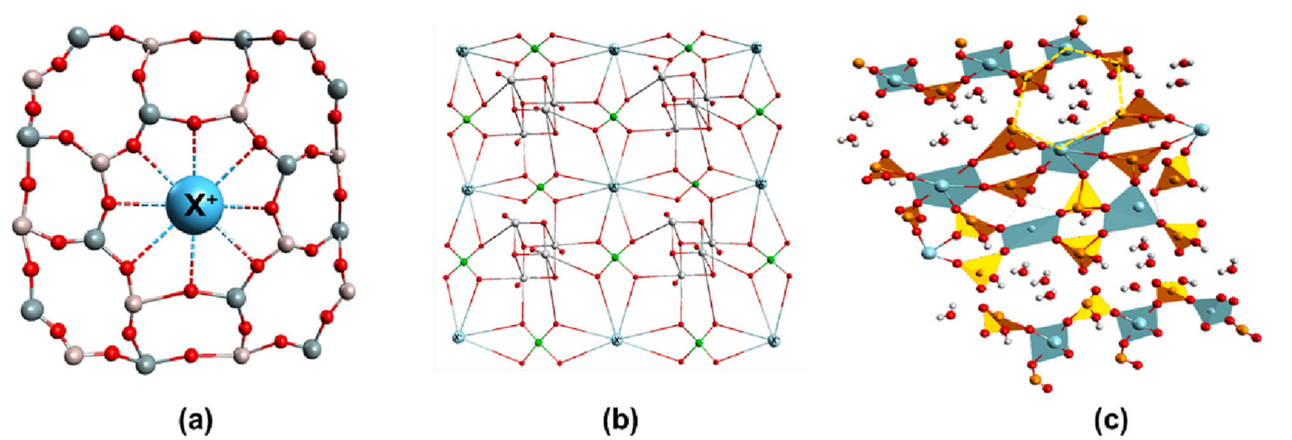


Fig. 9. Capture structure of (a) zeolite (b) titanium silicate (c) zirconium phosphate *(yellow dash hexagon denotes the pseudo-channel along [010] direction).*

competitive ions. Likewise, zeolite's performance for Cs is well established and therefore can also directly be reflective for Rb. It must be highlighted that the main ion exchange mechanism for zeolite is sodium-Rb exchange. In this regard, studies have emphasized the extremely low selective ion exchange capacity of zeolite in high saline brine/nuclear wastewater conditions. This is attributed to the high ion saturation effect that makes the exchange of Na from the material into a concentrated sodium water source highly challenging.

ii. Ammonium —Rb exchange

The ammonium-Rb^+ exchange mechanism has been explored in the case of ammonium-based adsorbent, AMP. In AMP adsorbent, ammonia in the centre of the crystal structure is reported to exchange with monovalent alkali metals, namely Na, K, Rb, and Cs [62,107,108]. According to previous studies, in a mixed concentration containing an equal concentration of all ions, the ammonium-alkali ion exchange follows the order of Cs > Rb > K > Mg > Na > Li indicating the suitability of using AMP for selective Rb recovery, alike Cs.

However, in a saline mixed brine solution, the AMP performance for selective alkali ions is sternly affected by the presence of Na, Ca, and NH_4. For instance, Park et al. [109] reported that the selective uptake of Cs onto polymer encapsulated AMP was reduced significantly by 90 % in saline brine containing a high concentration of Na as Na competes on the same sites to exchange with ammonium ions. Likewise, in another study, Mimura et al. [110] reported that the high performance of AMP for Cs uptake was due to the ion-exchange mechanism between NH_4 to Cs. However, it was decreased in the order of coexisting cations of H > Na > K > NH_4, following the reducing order of the size of hydrated ions.

iii. Potassium —Rb exchange

The ion exchange of potassium-Rb is predominantly displayed by various inorganic metal-based potassium hexacyanoferrate (KMFC, with M as transition metals such as Ni, Co, Cu, Fe) adsorbent. The mechanism for the ion-exchange reaction between the KMFC sorbent and Rb in an aqueous solution can be described by Eq. (2) [111]. Even though metal-based potassium hexacyanoferrate was initially developed for Cs recovery, they have also been used for Rb recovery [112–114].

$$K_2\left[MFe(CN)_6\right] + 2\,Rb^+ \rightarrow Rb_2\left[MFe(CN)_6\right] + 2\,K^+ \tag{2}$$

The selectivity of KMFC towards Rb in an aqueous solution can be explained through the ionic structures of the cation. Rb^+ has a similar hydrated radius with K^+, thereby favoring the replacement with K^+ on the surface and/or in the centre of the KMFC sorbent [115,116]. Meanwhile, the other cations such as Li, Na, and Ca have a larger hydrated radius, which follows the order $Ca^{2+} > Li^+ > Na^+ > K^+ \approx Rb^+ \approx Cs^+$ [117].

[54] Investigated the potential of potassium cobalt hexacyanoferrate (KCoFC, or CsTreat commercial) for the recovery of valuable metals from synthetic brine, namely Rb, Cs, and Li. The maximum adsorption capacity (q_{max}) of KCoFC for Rb (46.73 mg Rb/g) was found to be significantly higher than for Cs (32.36 mg Cs/g), while the uptake of other alkali ions, such as Li was negligible. It was indicated that KCoFC could be applied to recover Rb in an aqueous solution, besides the main competition with Cs (reduction of 21 % uptake capacity). Another study reported that KCoFC had a high sorption capacity for Rb (q_{max} = 100.10 mg/g) in comparison with other ions such as Na, Ca, and Li (<2 mg/g) [118]. In the solution with the simultaneous presence of these cations, the sorption of Rb decreased slightly. Only high Cs concentration (10 mg/L) reduced Rb uptake by around 20–25 %. However, given the insignificant presence of Cs in natural seawater and its related brine, Cs is not listed as a competing cation for the sorption process of Rb by KCoFC adsorbent under actual conditions. In another study, Naidu et al. [99] compared the performance of four different metal combinations of KMFC, including KCoFC, KCuFC, KFeFC, and KNiFC. KCuFC demonstrated better sorption capacity for Rb than the other metal based types, which was associated to the high negative surface charge (−30 to −32 mV) of KCuFC. Furthermore, the optimum pH range for the Rb uptake of KMFC was from 6 to 8, which is a direct match to the natural pH of seawater and brine [118].

Overall, Rb uptake by inorganic adsorbents has a vast potential and efficiency. However, limited studies in that field resulted in a massive squandering of the brine resources. More in-depth researches on material fabrication and operating conditions are crucial in the future to put the Rb recovery by adsorption method into commercial practice. Details of the studies carried out so far are presented in Table 8.

4.3.1.3. Encapsulation of ion exchange adsorbent. It should be emphasized that these inorganic ion-exchangers have a natural microcrystalline structure and a small or fine powder particle size, which leads to their limitation in practical application. This in turn, relates to the difficulty of the material recovery stage and hampers the hydraulic characteristics of the column system (i.e., loss of pressure, bed clogging, and filtration rate decrease) [99,119]. The materials could be modified to overcome these complications by immobilizing them with oxide materials or encapsulating the sorbents in polymer agents [62,119].

For example, Ye et al. [120] fabricated a composite spherical material (CSM) from AMP, sodium alginate (NaALG), and calcium chloride ($CaCl_2$). The results specified that the uptake process reached the equilibrium state quickly, which was within 2 h. The author also found that the sorption capacity decreased with the increase in ionic strength due to the ionic exchange competition in the solution.

Taj et al. [121] synthesized the polymethylmethacrylate (PMMA) supported iron (III) hexacyanoferrate(II) for the separation of alkali ions (Li, Rb, Sr, and Cs) from the waste solution. The sorbent exhibited a relatively high sorption capacity for Rb in the absence of competing ions, nonetheless, the material favored the other cations instead of Rb in the solution. Thus, the material was found only effective in treating nuclear waste solution, contrary to the Rb recovery purpose.

In another study, Krys et al. explored the performance of two different sorbents encapsulated in alginate beads, including AMP and hydrogen zirconium phosphate $Zr(HPO_4)_2$ (Phozir) [119]. The Rb sorption capacity of AMP was slightly higher than Phozir in the batch experiment (61.10 and 56.10 mg/g, respectively). As a result of this study, the maximum sorption capacity for both AMP and Phozir was significantly lower than the maximum exchangeable capacity (mmol eq/g) of the raw materials. This indicates that the sorbents are unable to absorb well because they are encapsulated in alginate beads. Furthermore, depending on the size magnitude, the smaller beads reached equilibrium within two hrs of contact time, while larger beads needed 4–6 h of experiment time. In the dynamic studies, with the uniform initial setup for column systems (C^o_{RB} = 25 mg/L, column internal diameter = 7 mm, flow rate = 42 mL/h, the mass of sorbent = 1 g), AMP-based material demonstrated better performance compared to Phozir modified adsorbent, with the sorption capacity of 36 and 22 mg/g (equivalent to the saturated bed volume of 3000 and 2000 BVs), respectively. The sorption capacity of these materials in dynamic operation was noticeably less than in static mode, which revealed that it is necessary to conduct intensive research on modifying the sorbents to optimize their sorption capacity.

Recently, Naidu et al. encapsulated KCuFC sorbent into an organic polymer (polyacrylonitrile, PAN) to obtain KCuFC-PAN. Due to the polymer mass in the encapsulation, the sorption capacity of KCuFC-PAN was significantly lower than the primary KCuFC (105.16 and 143.64 mg/g, respectively) [99]. On the other hand, if this inert content (30 % of dry weight) was not considered, the modified adsorbent exhibited an equivalent capacity compared to the original one. In dynamic studies, KCuFC-PAN showed an excellent adsorption property, with a high bed volume (13,400 BVs). The author also introduced some steps to improve the performance of KCuFC-PAN, by decreasing the flow rate or enlarging the particle size. That could be a potential research direction in the future for the application of KCuFC-PAN. Table 8 shows the performance of inorganic adsorbent.

Table 8
Performance of inorganic adsorbents for Rb recovery in aqueous solution.

Material	Exchangeable ion	Solution pH	Initial Rb concentration (mg/L)	q_{max} (mg/g)	Equilibrium time (hrs)	Ref.
Zirconium phosphate (ZrP)	H^+	7.80	5–80	3.32	24	[54]
KCoFC	K^+	7.80	5–80	46.73	24	[54]
KCoFC	K^+	7–80	5	100.10	15	[118]
KCuFC	K^+	7	5.13	143.64	24	[99]
KNiFC	K^+	7	5.13	122.27	24	[99]
KFeFC	K^+	7	5.13	52.15	24	[99]
Titanium silicate	H^+	2–12	4.3–213.80	478.80	240–288	[104]
AMP	NH_4^+	5–6	18.50	–	5	[62]
AMP-ZrP-SiO_2	NH_4^+ & H^+	5–6	18.50	–	5	[62]
PMMA-KFeCF	K^+	–	200	79.90	24	[121]
Polymer potassium copper hexacyanoferrate (KCuFC-PAN)	K^+	7	5.13	105.16	15	[99]
AMP–calcium alginate	NH_4^+	7	85–850	47.86	2	[120]
Ammonium molybdophosphate (AMP)	NH_4^+	3	10–500	61.10	72	[119]
Hydrogen zirconium phosphate	H^+	3	10–500	56.10	72	[119]

4.3.2. Metal-organic frameworks

Metal-organic frameworks (MOFs) are porous hybrid materials formed by metal ions and organic ligands [122,123]. MOF materials have many advantages, such as large surface areas, high pore volumes, low density, high chemical and thermal stability, and adjustable pore structure [122–125]. Due to these excellent properties, MOFs have wide applications, including catalysis [126,127], gas storage [128,129], adsorption [130,131], sensor [132,133], drug delivery [134,135], and others.

4.3.2.1. MOFs sorbents. Several researchers explored a group of MOFs named HKUST-1 for Rb recovery. Dai et al. [136] established the sorption performance of two materials, including HKUST-1 and HKUST-1, modified with phosphomolybdic acid (PMA). Promoting PMA into HKUST-1 helped enhance the sorption capability noticeably, with the q_{max} values of the original HKUST-1 and PMA@HKUST-1 being 76.31 and 99.47 mg/g, respectively. Furthermore, both materials took around 30 min to get saturated. However, it should be noted that the selectivity of both HKUST-1 and PMA@HKUST-1 for Rb in an aqueous solution was not considerable. Rb sorption capacities decreased to approximately half when competing cations (Na, K, and Cs) were present. In another study, Tain et al. coated polydimethylsiloxane (PDMS) onto the surface of HKUST-1 sorbent to form a new material, PDMS@HKUST-1 [137]. The motive for adding PDMS was not to increase the sorption capacity of the initial material but to increase the water-resistant characteristic, recyclability, and stability of HKUST-1. After five operational cycles, the sorption capacity of HKUST-1 decreased by 90 %, while the corresponding value for the modified HKUST-1 was almost unchanged.

Very few studies related to other MOF's have been published so far. Fang et al. coupled phenols (PH) with a MOF named MIL-101(Cr) to introduce a novel material PH@MIL-101(Cr) and conducted batch and column studies [138]. In this research, MIL-101(Cr) was a substrate for grafting PH on its surface. It was reported that for the batch studies, the raw MIL-101(Cr) had a maximum sorption capacity of 72.88 mg/g. The q_{max} of the modified material increased to 92.93 mg/g, which confirmed the association's effectiveness with PH. In addition, the Rb sorption of PH@MIL-101(Cr) reached the equilibrium state after 10–30 min, equivalent to the base HKUST-1 sorbents described above. The data obtained from the column study indicated that the PH@MIL-101(Cr) material was successfully applied in dynamic conditions for five consecutive operation cycles, with >90 % of the Rb^+ adsorbed. However, there are two things that need to be highlighted in this study. Firstly, the ratio of material/solution in the batch study was exceptionally high compared to other previous studies (0.50 g of sorbent per 5 mL solution, equivalent to 100 g/L). Conducting the adsorption experiments with high material content can lead to the saturation state taking place too quickly, making it difficult to determine the sorption properties accurately. Secondly, the effect of competing cations was not investigated, therefore, the applicability of the sorbent in the brine was not fully revealed.

4.3.2.2. MOFs combined with inorganic sorbents. Tian et al. modified Fe_3O_4 magnetic nanoparticles with sulfhydryl group (-SH). Then, HS-Fe_3O_4 was assembled with MIL-53(Al) to form a novel composite material HS-Fe_3O_4@ MIL-53(Al) [139]. The incorporation between HS-Fe_3O_4 and MIL-53(Al) promoted the Rb uptake capability and eased some limitations of the raw Fe_3O_4 particles. This material was found to have the equivalent Rb sorption capacity compared with the other reported MOFs. However, it showed poor selectivity towards Rb in the solution containing Na, K, and Cs. Recently, Wang et al. grafted AMP onto MOF-76(Sm) to generate AMP@ MOF-76(Sm) sorbent [140]. The fabricated material was effective within a wide range of solution pH (i.e., 6.0–10.0). Furthermore, 3.0 M NH_4NO_3 solution was found to be the most appropriate liquid for desorption. However, the effect of co-existing cations on the sorption performance of the material was only studied with the Na^+. Recently, Dai et al. [141] investigated the feasibility of combining potassium cobalt hexacyanoferrate (KCoFC) and ZIF-8 MOFs. The composited nanomaterial KCoFC@ZIF pointed out eight times higher Rb uptake with faster kinetics than the original inorganic sorbent KCoFC. In addition, KCoFC@ZIF maintained the high selectivity for Rb in complex seawater solution, and it also showed high stability in terms of Rb uptake performance after several operational cycles.

In conclusion, the MOFs and modified MOFs materials executed several excellent properties compared to the conventional ones. Firstly, they took less time (less than one hr) to reach the equilibrium state, which is much faster than other inorganic materials mentioned above (from several hrs to days). In addition, the MOFs can be easily adjusted to enhance their effectiveness. However, MOF materials have a number of disadvantages that prevent their widespread application in Rb recovery. Most of them are highly unselective towards Rb in the media with the presence of other conflicting ions (i.e., Na, K, and Cs). Furthermore, integrating other components into MOFs may also result in the inverse effect for the original sorbent. For example, Tian et al. [137] reported that increasing the optimum PDMS dose coating onto the raw HKUST-1 surface from 2 g PDMS/g HKUST-1 to 3 g PDMS/g HKUST-1 resulted in the partly blocking of the pores. This ultimately causes the sorption capacity to decline. Table 9 summarises various MOFs adsorbents used for Rb recovery in an aqueous solution.

4.3.3. Organic adsorbents

4.3.3.1. Ion exchange resins. Ion exchange resins (IER) are insoluble polymer compounds having a network structure holding exchangeable and active functional groups. They can capture the desired element

Table 9
Summaries of MOFs adsorbents for rubidium recovery in aqueous solution.

Material	Exchangeable ion	Solution pH	Initial Rb concentration (mg/L)	q_{max} (mg/g)	Equilibrium time (hrs)	Ref.
MIL-101(Cr)	–	7	0–100	72.88	0.50	[138]
PH@MIL-101(Cr)	H^+	7	0–100	92.93	0.50	[138]
HKUST-1	–	–	0–100	76.31	0.50	[136]
PMA@HKUST-1	–	–	0–100	99.47	0.50	[136]
HKUST-1	–	–	0–100	83.76	0.50	[137]
PDMS@HKUST-1	–	–	0–100	83.45	0.50	[137]
MIL-53(Al)	–	7	0–100	57.41	0.50	[137]
KCoFC@ZIF	K^+	7	5	1279.35	10	[141]
HS(50)-Fe_3O_4@MIL-53(Al)	H^+	7	0–100	86.22	0.50	[137]
AMP@MOF-76(Sm)	NH_4^+	7	42.80	43.94	0.83	[140]

based on ionic strength, temperature, adsorbent surface, the hydrated radius of elements, and pH values [142]. Fig. 10 demonstrates factors affecting and factors responsible for adsorption using resins. Guo et al. utilized three sodium-based resins (LSC-100, LSC-500, D001) for the adsorption of Li, Cs, K, and Rb. Among them, D001 was found to be more selective towards Rb, and the preference order for adsorption was reported as Cs > Rb ~ K > Li [26]. The author hypothesised that the electrostatic interaction between the resin and cation was critical in cationic adsorption. The electrostatic interaction was guided through the presence of functional groups (SO_3^- for D001, $N(CH_2COO\text{-})_2$ for LSC-100, and $NHCH_2PO_3^{2-}$ for LSC-500) that mediate the process of adsorption once they release related ions.

Besides electrostatic interaction, another factor that plays a significant role in the adsorption via resins is the hydration action of alkali metals. This effect plays a substantial role in determining the selectivity of metal ions during the separation process. A weakly hydrated cation will typically partition into the subaqueous phase more readily than one that is more strongly hydrated. The chemistry of adsorption for metal ions when held within the adsorbent's nanopores relies upon ionic hydration [143]. For a large hydrated radius, the cationic centre remains far away from the surface, leading towards lower electrostatic interaction between the cationic species and the adsorbent. However, the hydrated radius is not always a reliable predictor for ion selectivity. Hydration energy becomes an essential parameter in accurately explaining the mechanism. Cations with lower hydration energies can transit into other phases more readily due to their ease in shedding water molecules.

Magd et al. studied the ion exchange for Rb over sulfonated toluene phenol-formaldehyde sodium-based resin [144]. The Rb adsorption was found getting affected by higher pH values. The desirable adsorption was achieved at pH levels of 7–9. Additionally, a drop in ion exchange efficiency was observed with increased temperature. This is attributed to the fact that the change in enthalpy is inversely related to temperature. As a result, the free energy available for ion exchange declines, and the rate of adsorption decreases as well. Another type of resin is phenol formaldehyde (PF) resin with weak dissociating phenyl functional groups bonded to it. These resins favour extremely alkaline circumstances for alkali metal separation. When a mixture of alkali metals is present, cation exchangers based on PF resins have been extremely selective for Rb. This selectivity is ascribed to the exchange of ions directly taking place over the phenol groups. In reference to it, Shelkovnikova et al. evaluated the adsorption behavior of Rb over two PF macroporous resins (Amberlite XAD 761 and FFS-1.4/0.7) [145]. The macro pore structure offered better ion exchange stability against chemical agents. These two resins solely had phenol as their primary functional group. At the alkaline state with a pH value of 12, a subtle increase in sorption was observed. Also, the resin's transition from acid to alkaline conditions goes with a considerable rise in swelling along with a change in resin colour. The phenolic group's pK value was skewed by a higher temperature, which reduced its adsorption capacity. Under rising temperatures, sulphonated PF resins are discovered to be more selective for Rb than other phenolic groups. Wagn et al. fabricated $[NH_3CH_3]_4[In4SbS_9SH]$, a 3-D ion exchanger with a pyramidal form and micropores covering it [146]. The Exchange of Rb^+ with an organic amine was encouraged by the pores. According to experimental findings, pore size was detrimental to increased Rb^+ adsorption effectiveness.

Ion-imprinted polymers (IIP) have evolved as a result of scientific advancements, improving their ability to promote Rb adsorption. For imprinting ions through polymerization into the polymer matrix, an appropriate ligand, primarily crown ether, is utilized. Compared to ligands having open-chain molecules, cyclic crown ether as ligands offers superior stability to the metal ion complexes. Fig. 11 discusses the ion imprinting technology's mechanism. According to Lewis's acid-base theory, the interaction between the functional monomer and ligand

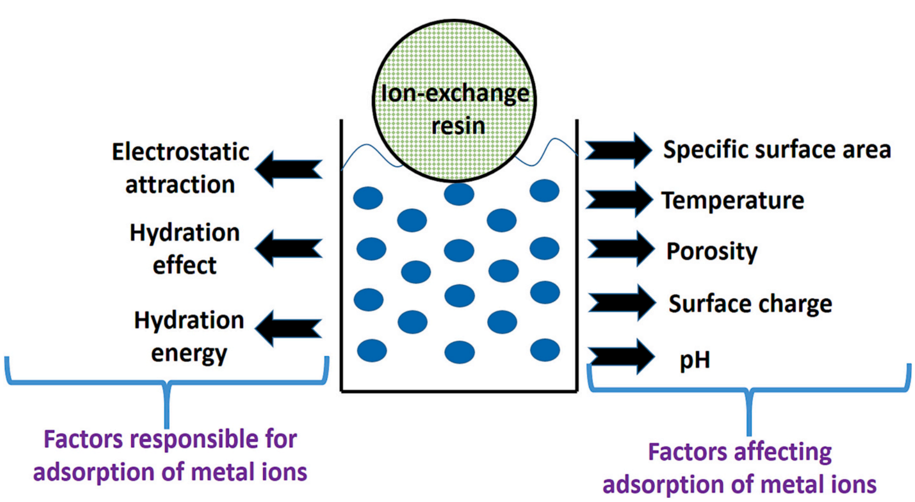


Fig. 10. Factors affecting and factors responsible for adsorption using resins.

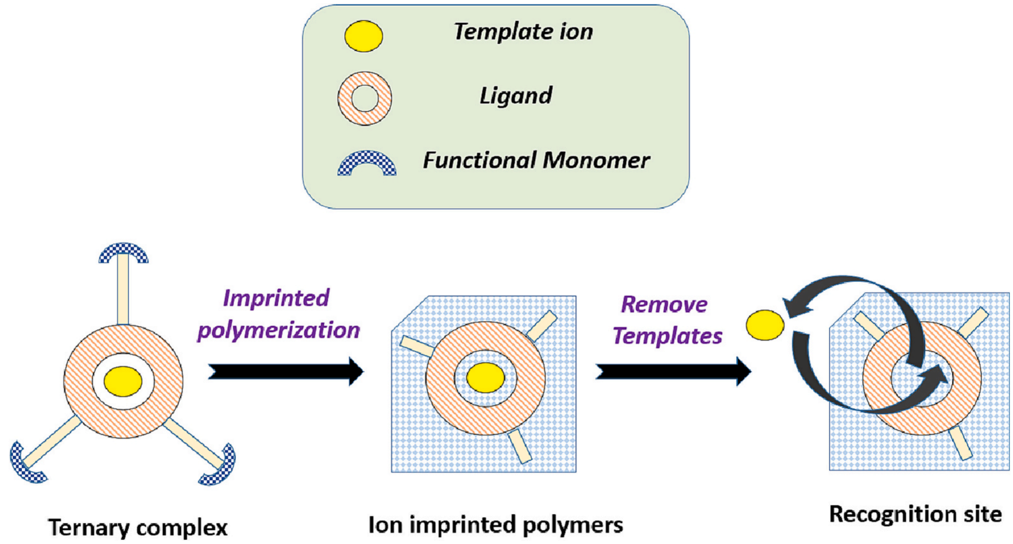


Fig. 11. Ion imprinting technology mechanism. The figure depicts the formation of ion-imprinted polymers, converting from the ternary complex through imprinted polymerization.

determines binding site stability and increases recognition ability. Later, this imprint ion is removed from the matrix through mineral leaching. Template-shaped cavities are created by ion imprinting, and imprinting polymers are subsequently created by copolymerizing functional and cross-linking polymers. Various IIPs developed regarding Rb recovery are discussed below.

Hashemi developed nanoparticles-based IIP utilizing dibenzo-21-crown-7 as a ligand [147]. This ligand proved useful in applications requiring cation selectivity and non-polar liquid solubility. Over a variety of competing metal ions, the newly created sorbent demonstrated a strong selectivity towards the Rb. The results showed that repeated use of this adsorbent did not cause any degradation in its adsorption capacity. Liu et al. utilized benzo-12-crown-4-ether as a ligand and fabricated Rb(I) IIP [29]. This adsorbent showed maximal sorption capacity at an alkaline pH value of 9.5. Also, the prepared IIP reflected better selectivity ratios (>1) and improved re-generation and re-usability. Zheng et al. prepared a dual-functional mesoporous film for the simultaneous adsorption of Li and Rb [148]. The desired porous film was fabricated utilizing a bio template made up of cellulose nanocrystalline (CNC). A layer of Rb and hydrogen manganese oxide (HMO) was then grafted to simultaneously enrich Rb and Li, respectively. The obtained film exhibited the best adsorption capacity at neutral pH. Ionic sieving and ion imprinting were vital in attaining selectivity during Rb and Li adsorption. Xu et al. utilized porous silica to fabricate hierarchical mesoporous material through a dual template technique for Li and Rb adsorption [149]. Surface characterization showed the formation of a good number of long channels responsible for adsorption throughout the porous adsorbent.

The ion-imprinting polymeric compounds are challenging to separate once dissolved in a solution. This is ascribed to their fine powder nature. Magnetic polymeric nanoparticles help in providing a solution to this problem. Firstly, it acts as a solid support, and secondly, it can be easily removed by applying magnetic force. Xu et al. prepared a magnetic ion imprinting polymer (IIP), where $Fe_3O_4@SiO_2$ was utilized as a supporter and crown ethers served as monomers for Rb removal from salt lake brine [150]. Jiang et al. applied graphene oxide functioning as a carrier for grafting magnetic nanoparticles on its surface through the co-precipitation method [151]. The stability of magnetic particles was achieved using polydopamine as a coating material. The best adsorption result was attained at an alkaline condition and offered the potential for reusability.

4.3.3.2. Supramolecular adsorbents. Crown ethers and calixarenes have some astounding selectivity for Rb^+ and by the interaction of the phenolic oxygen groups with the polyether chain, they combine to produce derivatives of the calyx crown [152]. Through noncovalent interactions, including as p-p and CH-p interactions, they have been acknowledged as hosts for a variety of guest metals. Calixarenes are a member of the third-generation supramolecular compounds and are proficient in producing a wide range of derivatives. Calixarenes macrocyclic phenolic rings serve as a molecular scaffold to which functional groups can be attached. Through chemical grafting or polymerization, they can combine with porous materials. An adsorbent is reported in the literature, which was prepared after functionalizing the porous surface of graphene oxide using thiacalix[4]arene (TC4A) through polymerization and esterification [152]. At lower pH levels, this adsorbent offered greater adsorption capability reaching a maximum value at pH 4–5. Several functional groups with oxygen atoms were reported on the porous surface. However, the behavior of the adsorbent varies under varying pH conditions. An extremely acidic environment caused oxygen atoms protonation. However, extremely basic circumstances resulted in competitive adsorption. Zhang et al. employed silica as a porous material for impregnating calixarene over it and the affinity was enhanced using methyloctyl-2-dimethylbutanamide (MODB) [153]. These adsorbents were slightly low selective under additional ions.

MOF, in conjunction with calixarenes paved a way to improve adsorption selectivity [154]. The MOF offers a structure with organized porosity and a greater specific surface area compared to other porous materials. Recently, some organometallic functionalization framework materials create MOFs and related materials composites have shown high adsorption capacity for some radioactive metals. In this reference, MOF based adsorbent Ui0–66@iPCC5 has been reported [154]. Lyu et al. prepared a solid-state supramolecular adsorbent consisting of (XAD-7), a polar haulier, and iPCalixC5X for selective adsorption of Rb [86]. Table 10 shows the performance of various ion exchange and supramolecular based adsorbents.

4.3.4. Others

In recent years, the phase emulsion process has been applied for the adsorption of Rb along with Cs and for other radioactive elements [155]. The process includes a dispersing phase, a continuous phase, and a stabilizer (mostly surfactants) for maintaining emulsion stability. An emulsion template can generate porous polymers with improved permeability, amplified surface area, and reduced density. Adsorbents

Table 10
Adsorption performance of supramolecular and ion exchange resin for rubidium ion.

	Operating parameters			Optimum conditions			
Adsorbents	Adsorbent dosage (*g/L*)	Temperature (*K*)	C0 (*mg/L*)	Contact time (*min*)	pH	Maximum adsorption capacity (*mg/g*)	Ref.
Supramolecular							
TC4A-GO	0.20		5–600		4–5	164.47	[152]
iPCC5@UiO-66		298		180	–		[152]
iPCalixC5X		298		120	–	158.74	[86]
Calix-MODB/silica		298		180	–		[153]
Ion exchange resin							
Rb/Li-IHPS	1	298	1		7	0.14	[149]
Rb/IIP- 18C6	2	299	5–600	6	9.80	213.08	[143]
D001	30	298	854.67	30		42.73	[26]
			213			12.82	
LSC-100	30	298	854.67	30		35.89	[26]
			213			11.11	
LSC-500	30	298	854.67	30		35.89	[26]
			213			10.25	
DIMF	1	300	50	100	7	6.21	[148]
Magnetic silica IIP	1	298	1		7	0.09	[150]
MIIP	0.10	299	400		10	186	[151]

made via pickering emulsion have flexibility to adjust their porosity behavior either externally (by altering pH and temperature) or internally (by altering the ratio of the stabilizer phase to the dispersion phase). Among various emulsion templates utilized, high internal phase emulsions (HIPEs) and medium internal phase emulsions (MIPEs) have been significant. They have resulted in the formation of polymers that display great adsorption values [156–158]. The addition of surfactants as a stabilizer during HIPEs formation is expensive and damaging to the environment. Recently, research has seen progress in the usage of solid particles as a replacement for surfactants. In this reference, yeast is reported as a stabilizer after treating with caustic (P-yeast) as a pretreatment [155]. This inherited amphiphilic character (presence of functional groups like amino, carboxyl, and hydroxyl) and stabilizing properties to yeast.

A novel adsorbent was found and fabricated using acrylic acid and yeast emulsion. The adsorbent demonstrated better regeneration capability and improved reusability up to five cycles of adsorption-desorption. However, features like magneticity and photocatalysis can be induced by altering the types of nanoparticles used in the emulsion as stabilizers. Zhu et al. fabricated spheres having porous magnetic nature through graft polymerization between hydroxypropyl cellulose (HPC) and acrylic acid (AA) along with Fe_3O_4 nanoparticles [159]. The Fe_3O_4 particles were modified using orthosilicate and amino silane groups. In comparison to previously obtained results, the as-prepared magnetic adsorbent demonstrated improved adsorption and attained its adsorption capacity in 15 min. In one study, pickering MIPEs with magnetic yeast, chitosan as a stabilizer, and soybean oil as a dispersed phase provided a tunable porous magnetic adsorbent [160]. Similarly, magnetic porous spheres were fabricated through the linking of acrylic acid on HPC using HIPEs emulsion [159]. Apart from yeast, Attapulgite (APT) along with chitosan (APT/CTS) was applied as a stabilizer and a 3-D tunable porous adsorbent were fabricated using polyacrylamide [161]. It was concluded that the electrostatic attraction and complexing process between adsorbent and metal ions is the primary reason for high and rapid adsorption rates.

Pressure driven membranes have been acknowledged in recent years because of their role in various fields, including desalination, separation of metal ions, and pollutants removal in water. Sun et al. prepared a mixed matrix membrane using AMP/PSF utilizing the non-solvent induced phase inversion (NIPs) method [162]. The Rb adsorption over the membrane becomes possible due to the hydrophilic AMP that migrated towards the water interface during NIPs to diffuse Rb particles rapidly. Additionally, AMP doping increased the hydrophilicity in general, which boosted stability and speed up the pace of phase separation. This in turn, improved the pore size and hence increased the flux rate.

One of the studies employed the foam separation method for selectively removing alkali metals from anionic surfactants. Once the charged metal ions were adsorbed onto the ionic aliphatic chain, they were removed utilizing air bubbles [163].

Recently, protonated molecular tunnels made from manganese oxide-based material (doped with cobalt and nickel) were utilized as an electrode for Rb recovery [164]. Due to the reduction of manganese into its lower state, the adsorption performance under electrochemical circumstances was enhanced. Protonated manganese oxide doped with nickel was found to be efficient and optimum. The higher stability and crystallinity make these materials a good candidate for Rb enrichment. The adsorption performance of adsorbents discussed in this section is illustrated in Table 11.

5. Economic aspect of rubidium recovery

The economic benefit of Rb recovery is reflected not just in the revenue it will generate, but also in the fact that it will help considerably lower the brine toxicity levels. The revenue potential vis a vis total costs related to the recovery process need to be assessed and will heavily influence the economic viability of Rb recovery from seawater brine. Due to the expanding industrial Rb application, as discussed in the preceding sections, demand for Rb is escalating. Along with the increasing demand and potential uses, the price is also gradually increasing. Reverse osmosis and other desalination facilities yield concentrated brine that can be used to extract Rb, providing abundant Rb metal for industrial use and reducing the burden of land-based mining. Considering the concentration of Rb present in the brine as 0.18–0.20 mg/L, the amount of Rb that can be extracted through recovery processes and the monetary benefit it will offer is described below Table 12.

5.1. Economic evaluation of different processes

For the purpose of treating brine for Rb recovery, the precipitation method might be economical. The process substantial expense is incurred by the materials and chemicals utilized in it. Most of these chemicals are in cost ranges of approximately USD 37–800(INR 3000–6000) (price as per Merck and Sigma-Aldrich). Despite the fact that these compounds are simple to employ, the reaction time is generally longer. While some of these compounds exhibit rapid precipitation, they are either expensive or involve a complex purifying process. Sometimes, when using a specific precipitating agent, recrystallization becomes necessary. The creation of secondary contaminants is another

Table 11
Adsorption performance of other adsorbents for rubidium ion.

	Operating parameters			Optimum conditions			
Adsorbents	Adsorbent dosage (*g/L*)	Temperature (*K*)	C0 (*mg/L*)	Contact time (*min*)	pH	Maximum adsorption capacity (*mg/g*)	Ref.
AMP/PSF mixed matrix membrane	–	298	50	–	7	98.40	[162]
MPS	0.80	303.15	500–5000	15	6	310	[159]
H-OMS-5	–	298	3 (g/L)		7	390.30	[164]
H-Co-OMS-5	–	298	3 (g/L)		7	405.70	[164]
H-Ni-OMS-5	–	298	3 (g/L)		7	481.70	[164]
Mag molecular	0.01	298	363.63		7	162.60	[165]
IMO	1	297–299	0.125–2.50 mmole/L	15	4	0.17 mmole/g	[166]
IMO (MBI)	1	297–299	0.125–2.50 mmole/L	18	4	0.29 mmole/g	[166]
IMO (MTZ)	1	297–299	0.125–2.50 mmole/L	30	4	0.30 mmole/g	[166]

Table 12
Economic analysis of Rb recovery from brine [53].

Considering the minimum concentration of Rb as 0.18 mg/L in brine and process recovery efficiency as 90 %			
	Rb that can be extracted from brine (Kg)	Cost of Rb (USD) (Cost of Rb/Kg in USD X Rb extracted)	Cost of Rb (INR) (Cost of Rb/Kg in INR X Rb extracted)
Per day	0.15	2211.33	174,856.11
Per month	4.50	66,340.08	5,245,683.29
Per year	54.08	796,081.06	62,948,199.54

Considering the maximum concentration of Rb as 0.20 mg/L in brine and process recovery efficiency as 90 %			
	Rb that can be extracted from brine (Kg)	Cost of Rb (USD) (Cost of Rb/Kg in USD X Rb extracted in Kg)	Cost of Rb (INR) (Cost of Rb/Kg in INR X Rb extracted in Kg)
Per day	0.18	2730.04	217,112.27
Per month	5.56	81,901.34	6,513,368.18
Per year	66.76	982,816.12	78,160,418.21

factor that renders this method impractical. Removing these secondary products will require additional treatment utilizing other methods such as advanced oxidation and UV radiation. Hence, overall this method has certain barriers to economic Rb recovery.

For the solvent extraction method, the cost of extractants, solvents, and solvent extraction equipment are all capital expenses associated with the process. A significant quantity of chemicals and solvents are required on a large industrial basis. Looking into the work of Liu et al. who used crown ether for Rb recovery, can provide essential insight [90]. They utilized 0.26 g of crown ether for extracting Rb from solution containing 0.1 mg Rb. If this work is to be taken as a reference, then extracting Rb from brine containing 150,226.66–185,465 mg Rb will require around 400 Kg of crown ether and the average cost is around USD 5/g (INR 2000). This huge amount of investment will be more expensive in comparison to the revenue generated through Rb recovery. Additionally, handling these chemicals frequently will raise maintenance costs. This process also involves the removal of certain co-existing ions as a pre-treatment. Besides, techniques like centrifugal separation, scrubbing, and stripping are also employed, increasing the overall cost of the entire process.

Adsorption offers an advantage over the other known Rb recovery methods. This technique has a high level of selectivity and increased efficiency. In comparison to other processes, this process has reduced operating and capital costs. It can operate over a wide pH range, so it won't requisite any additional pre-treatment units and chemicals for pH modification. Moreover, it is simple to use other hybrid approaches to produce pure water and simultaneously recover metals.

As such significantly less research has been dedicated to the economic aspect of various recovery processes for extracting elements from seawater. Evidently, the Rb recovery from seawater brine warrants us to investigate the economic aspect of the process further so that the cost of recovery can also be completely quantified.

6. Conclusion and outlook

The rising demand and escalating price of rubidium (Rb) as an essential element in biomedical, specialty glass, electronics, solar-cells, atomic clocks, and pyrotechnics entails instant attention enabling long-term recovery of Rb from alternative sources to maintain the supply and demand equilibrium in future markets. Additionally, the large reserves of Rb in seawater and other natural water sources is propelling the implementation of potential Rb recovery processes from these sources. For Rb recovery the use of precipitation, solvent extraction, adsorption, membrane-based methods, phase emulsion processes, foam separation, and electrochemical methods are the major technologies that have been employed. As discussed, conventional technologies such as precipitation and solvent extraction methods will not be able to fulfill the rising demands of Rb.

The solvent extraction process includes extractants that can extract Rb either under highly acidic or basic conditions. It is essential to explore advanced extraction systems that can work well for neutral conditions. For Rb recovery, the focus of researchers in recent years has been on phenols, substituted phenols, and crown ethers. So far, specific consideration has been given to 18-crown-6, benzo crown ether, dicyclohexane-18-crown-6 crown ethers, and t-BAMBP.

In order to remove Rb from aqueous and brine solutions, adsorption has received more attention. In this review, several kinds of adsorbents, including organic, inorganic, MOFs have been explored thoroughly. As adsorbents are constantly being fabricated and utilized for Rb recovery, there has been increased recognition of the advancement of adsorption mechanisms and modification techniques. All the as-discussed adsorbents have reflected efficient recovery efficiency for Rb. However, these adsorbents differ in selectivity and their performance when combined with other agents.

6.1. Performance under competing ions

Inorganic adsorbents are more efficient than organic adsorbents because of their radio stability. Among all inorganic adsorbents, KMFCs, AMP, and MOFs hold astounding and promising selectivity for Rb^+. Out of all the discussed metals, potassium metal hexacyanoferrates, KCuFC has demonstrated high selectivity and sorption capacity in saline condition over major ions in seawater. This is due to the presence of a cubic crystal structure and a higher negative surface charge. The selectivity of zirconium-based adsorbents has also been confirmed. This results from the fact that the structure's spacing and interlayers can be altered according to the target element. Other than that, maintaining selectivity for titanium silicates and zeolites is challenging. Despite their microcrystalline structures, none of them can be used in columns. Therefore a variety of modifications, including grafting, immobilization, and magnetic modification, have been utilized.

Organic adsorbents selectivity depends on factors such as electrostatic interaction, presence of functional groups, hydration action of alkali metals, hydration energy, pH conditions, and pore size. Based on these factors, cation exchangers utilizing PF resins have been extremely selective for Rb in the presence of competing ions. PF ion-exchange resins as an organic adsorbent have received significant attention due to their extraordinary adsorption capacities under varying pH and temperature values. Ion-imprinted polymers (IIP) have also evolved as a result of scientific advancements and have shown drastic improvement in their ability to promote Rb adsorption. Although more comprehensive research is required to comment on selectivity within the presence of competing ions. Additionally, crown ethers and calixarene create supramolecular structures useful for Rb adsorption. Other methods, including phase emulsion process (HIPEs and MIPEs), pressure-driven adsorptive membrane-based methods, and foam separation, are also briefly introduced in this review to impart a broad understanding of the topic. However, more experimental data are required to investigate the performance of these adsorbents under the presence of competing ions.

6.2. Performance with encapsulation agent

The performance of adsorbents can also be affected when combined with encapsulating agents. MOFs show promising capacity by enhancing the available surface area for ion diffusion, thereby increasing Rb uptake at faster kinetics. The process of encapsulating the sorbents with polymer and alginate bead affected the uptake ability of sorbents. However, the smaller beads reached adsorption equilibrium faster in comparison to larger beads.

7. Future recommendations

1. For the precipitation method, more emphasis should be given to the development of a suitable precipitant for selective Rb^+ separation and efficient recovery.
2. Considering the high efficiency of t-BAMBP for Rb recovery through solvent extraction, further work needs to be done to reduce its cost as well as develop more environmentally benign diluents.
3. So far, researchers have employed batch extraction method for Rb. Future research should focus more on integrating various technologies.
4. The sorption capacity of adsorbent materials in dynamic operation is less noticeable than in static mode, which revealed that it is necessary to conduct intensive research on modifying the sorbents to optimize their sorption capacity under dynamic conditions.
5. The performance of potassium metal hexacyanoferrates such as KCoFC can be significantly enhanced by grafting them with MOFs.
6. MOFs for application in the recovery of Rb needs to be explored in more detail, specifically varied conditions of synthesis to identify the most optimal and suitable materials to avoid pore blocking.
7. Further information is solicited on the desorption process. More emphasis should be given to finding a suitable desorbing agent for Rb purification after adsorption.
8. While the studies mentioned in this review state promising results for Rb recovery, most of them obtained these results for synthetic or simulated brine solutions where the outcomes could substantially differ from what would occur with actual brine. Hence, the performance of these adsorbents with actual brine needs to be evaluated.
9. Furthermore, a comprehensive techno-economic feasibility analysis for various Rb recovery methods is crucial.
10. Future research should more explicitly concentrate on feasibility analysis and expanding lab-scale technology to suit industrial-level usage.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

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